

Recommender Systems and Self-Preferencing

Chiara Margaria

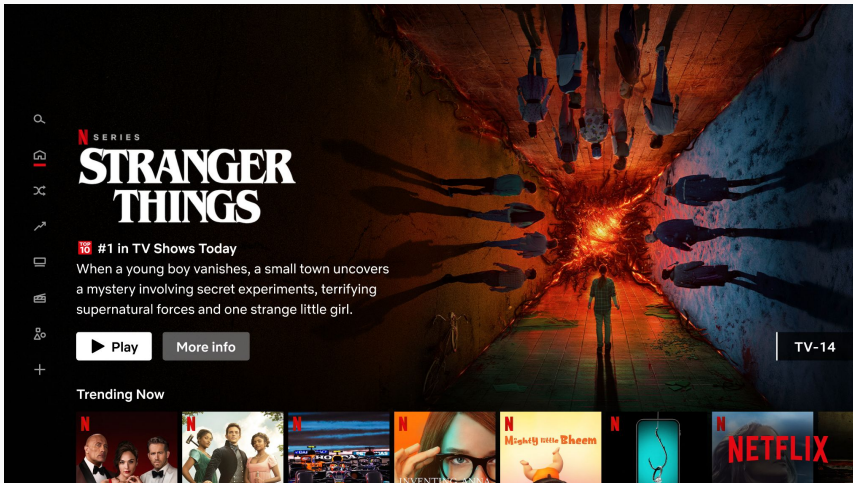
Boston University

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Introduction

- ▶ Digital platforms have access to a vast amount of data on users and products.
- ▶ A recommender system is the tool used by platforms to aggregate this information and guide users toward the products they may be most interested in.

Introduction

The banner features a dark, atmospheric scene of a group of children standing in a long, dimly lit hallway. The walls are covered in graffiti, and the floor is made of wooden planks. A bright, glowing orange light emanates from the end of the hallway, where a single child is walking away from the viewer. The overall mood is mysterious and slightly ominous.

N SERIES

STRANGER THINGS

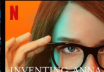
#1 #1 in TV Shows Today

When a young boy vanishes, a small town uncovers a mystery involving secret experiments, terrifying supernatural forces and one strange little girl.

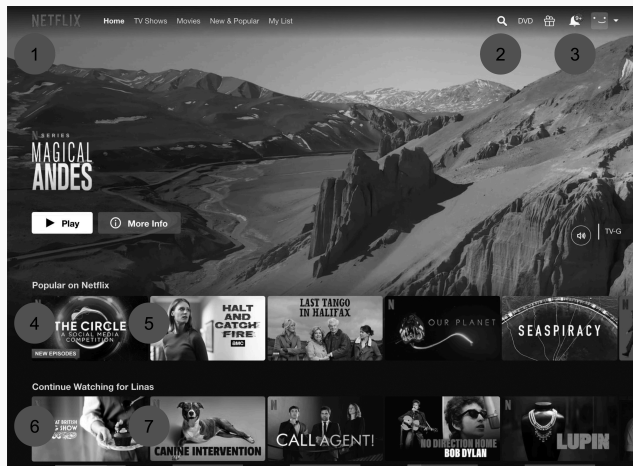
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TV-14

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Introduction

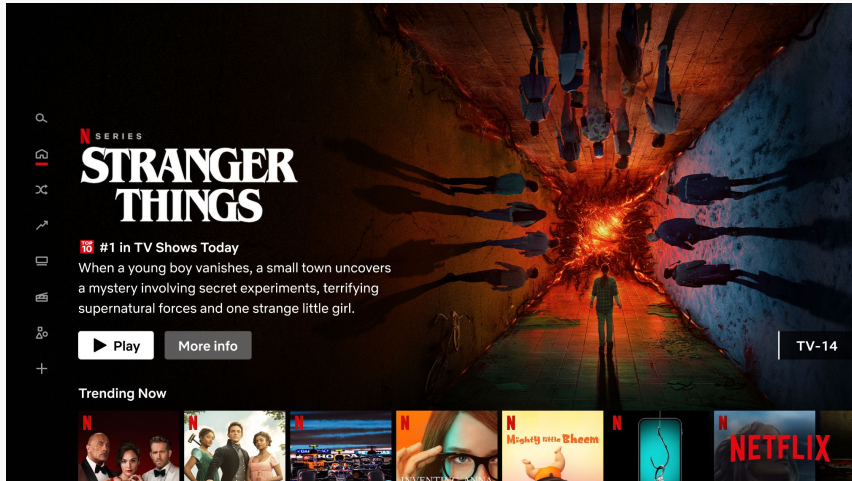


Source: Steck, H., L.Baltrunas, E.Elahi, D.Liang, Y.Raimond, and J.Basilico, “Deep Learning for Recommender Systems: A Netflix Case Study,” *AI Magazine*, 2021, 42, 7–18.
(Netflix homepage where we highlighted various recommendation tasks.)

Introduction: Recommender Systems as Experts

- ▶ Much like experts, recommender systems must build a reputation for providing recommendations that are “informative enough” to be trusted by users.
- ▶ Unlike experts, they rely on user data to improve their recommendations, so credibility must be built gradually.
- ▶ And like experts, they may be systematically biased toward certain products.

Introduction: Self-Preferencing



Introduction: Self-Preferencing

- ▶ A regulatory concern: the practice of self-preferencing.

The gatekeeper shall not treat more favourably, in ranking and related indexing and crawling, services and products offered by the gatekeeper itself than similar services or products of a third party. The gatekeeper shall apply transparent, fair and non-discriminatory conditions to such ranking.

Regulation (EU) 2022/1925, Digital Markets Act

- ▶ Waldfogel (2024) documents a sharp change in Amazon's self-preferencing following its designation as a "gatekeeper" under the DMA: Amazon's own products' rank advantage fell from roughly 30 to about 20 positions.
- ▶ Detecting self-preferencing bias is challenging because differential treatment may simply reflect legitimate quality or match-value differences.

What I Do

- ▶ Build a dynamic model of a recommender system that mediates social learning (Kremer, Mansour, and Perry (2014) and Che and Hörner (2018)).
- ▶ Characterize the optimal recommendation policy under different objectives and its implications for learning dynamics.
- ▶ Show how the objective (consumer welfare vs sellers' sales) shapes the time path of exploration and market shares.

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Not in the Model

- ▶ Prices.
- ▶ Horizontal differentiation.

Outline

1. Model
2. Dynamic Incentive Problem
3. The Optimal Policy in a Special Case
4. Main Result
5. Implications for Self-Preferencing
6. Conclusion

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Model

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- ▶ A long-lived designer and a constant unit flow of myopic agents.

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- ▶ Time is continuous, $t \geq 0$.
- ▶ A long-lived designer and a constant unit flow of myopic agents.
- ▶ At each time:
 1. the designer provides a private recommendation to each of the time- t agents;
 2. each agent chooses between consuming product A , product B , or neither.

Model: Designer

- ▶ The designer is forward-looking and commits to a recommendation policy.
- ▶ By the revelation principle, I can restrict attention to direct action recommendations.
- ▶ At any time and after any private history, the designer sends a private recommendation to each of the time- t agents, that is, she recommends
 - product A to a fraction a_t ,
 - product B to a fraction b_t ,
 - the outside option to a fraction $1 - a_t - b_t$.

Model: Learning

- ▶ The two products are ranked in quality: one of the two products $\theta \in \{A, B\}$ is better than the other for all consumers.
- ▶ If a fraction a_t of agents consume product A , and a fraction b_t consume product B over an interval of length dt , the designer (privately) observes:

	$\theta = A$	$\theta = B$
positive feedback about B	0	$b_t dt$
no feedback	$1 - a_t dt$	$1 - b_t dt$
positive feedback about A	$a_t dt$	0

Model: Learning

- ▶ At time t , the designer's belief that $\{\theta = A\}$ is either:
 - 1 if she has received some positive feedback about product A ;
 - 0 if she has received some positive feedback about product B ;
 - p_t , where

$$\dot{p}_t = -(a_t - b_t)p_t(1 - p_t).$$

Model: Designer's Initial Information

- ▶ The designer has access to some prior information about θ .

designer's initial signal	$\theta = A$	$\theta = B$
positive feedback about A	0	$1 - B_0$
no feedback	B_0	B_0
positive feedback about B	$1 - B_0$	0

- ▶ Assume $1 - B_0 \ll 1$, so the initial information advantage of the designer is small.
- ▶ For example, the designer has access to a mass $\mathcal{B}_0$ of beta-testers, and $B_0 = \exp(-\mathcal{B}_0)$.

Model: Designer's Initial Information

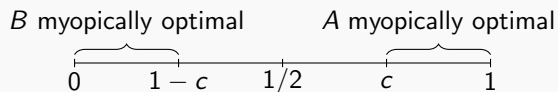
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designer's initial belief	$\theta = A$	$\theta = B$
1	0	$1 - B_0$
p^0	B_0	B_0
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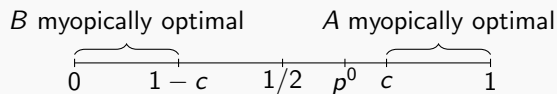
Model: Agents

- ▶ Agents face an opportunity cost of consumption $c \in (1/2, 1)$.
- ▶ Utility from consuming product A is $\mathbf{1}\{\theta = A\}$.
- ▶ Utility from consuming product B is $\mathbf{1}\{\theta = B\}$.



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- ▶ At time 0, agents attach prior $p^0 = \Pr\{\theta = A\} \in [1/2, c)$ to product A being the best product.

Designer's Problem

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Designer's Problem

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- ▶ For a benevolent designer, $\omega_A = \omega_B = 0$.

$$\mathbb{E} \left[\int_0^\infty \left((\mathbf{1}\{\theta = A\} - c) a_t + (\mathbf{1}\{\theta = B\} - c) b_t - (1 - c) \right) dt \right],$$

- no discounting;
- strong long-run average criterion.

Designer's Problem

- ▶ The designer's objective is to maximize a weighted sum of agents' welfare and the revenues of the sellers, with weights 1, ω_A , and ω_B , respectively.
- ▶ In general

$$\mathbb{E} \left[\int_0^\infty \left((\mathbf{1}\{\theta = A\} - c + \omega_A) a_t + (\mathbf{1}\{\theta = B\} - c + \omega_B) b_t - (1 - c + \omega_\theta) \right) dt \right],$$

- no discounting, strong long-run average criterion;
- normalize to 1 each firm's revenues for serving all agents;
- assume $\min\{\omega_A, \omega_B\} \geq 0$ and $\max\{\omega_A, \omega_B\} < c$.

First-Best

As in Klein and Rady (2011), the designer would like agents to consume

- ▶ product A only when $p_t > p^*$;
- ▶ product B only when $p_t < p^*$;
- ▶ product A and B in equal proportion when $p_t = p^* := 1/2 + (\omega_B - \omega_A)/2$,

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Recommendation Policy

- ▶ Without loss, the designer's recommendation policy can be described by two functions, $(\alpha_t)_{t \geq 0}$ and $(\beta_t)_{t \geq 0}$, i.e., the designer recommends:
 - product A with probability 1 if $p_t = 1$;
 - product B with probability 1 if $p_t = 0$;
 - product A and product B with probability α_t and β_t , respectively, if $p_t \in (0, 1)$.

Incentive Problem

- ▶ At $t > 0$, the informational advantage of the designer depends on the products she recommended in $[0, t)$.
- ▶ For example, at time t , an agent is willing to follow a recommendation to consume product B only if the posterior likelihood ratio exceeds $c/(1 - c)$, i.e.,

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$$\frac{1 - p^0}{p^0} \cdot \frac{B_0 \exp\left(-\int_0^t \beta_s ds\right) \beta_t}{B_0 \exp\left(-\int_0^t \alpha_s ds\right) \beta_t} \geq \frac{c}{1 - c}.$$

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- ▶ The more the designer has recommended product B in the past, the more credible her recommendation to consume product B is at time t .

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- ▶ The more the designer has recommended product B in the past, the more credible her recommendation to consume product B is at time t .

Incentive Constraint for Product B

- The IC for product B puts an upper bound on the share of agents that can be recommended product B at time t :

$$\beta_t \leq \bar{\beta}(\ell_t, B_t) := \begin{cases} 1, & \text{if } \ell_t \leq \frac{1-c}{c}, \\ \min \left\{ \frac{1-c}{c} \cdot \frac{(1/B_t - 1)}{\ell_t - (1-c)/c}, 1 \right\}, & \text{if } \ell_t > \frac{1-c}{c}. \end{cases}$$

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where

$$\ell_t := \frac{p^0 B_0 \exp \left(- \int_0^t \alpha_s ds \right)}{(1 - p^0) B_0 \exp \left(- \int_0^t \beta_s ds \right)}$$

likelihood ratio

$$B_t := B_0 \exp \left(- \int_0^t \beta_s ds \right)$$

Pr no feedback conditional on $\theta = B$.

Incentive Constraint for Product A

- Similarly, the IC for product A puts an upper bound on the share of agents that can be recommended product A at time t :

$$\alpha_t \leq \bar{\alpha}(\ell_t, B_t) := \begin{cases} 1, & \text{if } \ell_t \geq \frac{c}{1-c}, \\ \min \left\{ \frac{\ell_0 - B_t \ell_t}{B_t(c/(1-c) - \ell_t)}, 1 \right\}, & \text{if } \ell_t < \frac{c}{1-c}. \end{cases}$$

where

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likelihood ratio

$$B_t := B_0 \exp \left(- \int_0^t \beta_s ds \right)$$

Pr no feedback conditional on $\theta = B$.

Dynamic Problem

- ▶ At each $t \geq 0$, the designer chooses $\alpha_t \in [0, 1]$ and $\beta_t \in [0, 1]$ subject to:
 1. IC for product A: $\alpha_t \leq \bar{\alpha}(\ell_t, B_t)$,
 2. IC for product B: $\beta_t \leq \bar{\beta}(\ell_t, B_t)$,
 3. capacity constraint $\alpha_t + \beta_t \leq 1$.

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The Optimal Policy in a Special Case

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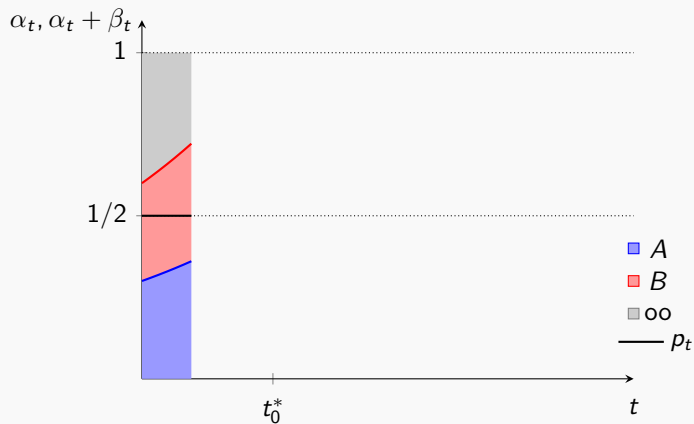


Figure: $c = 5/8$, $B_0 = 5/6$.

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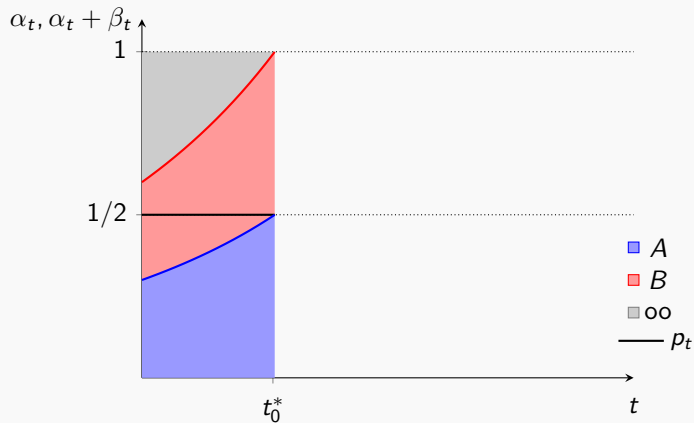


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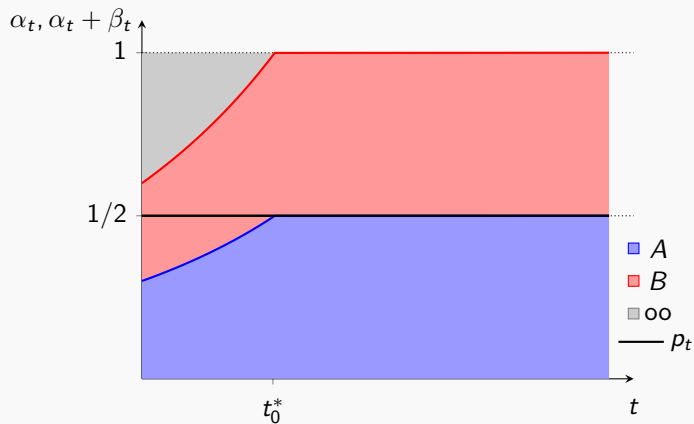


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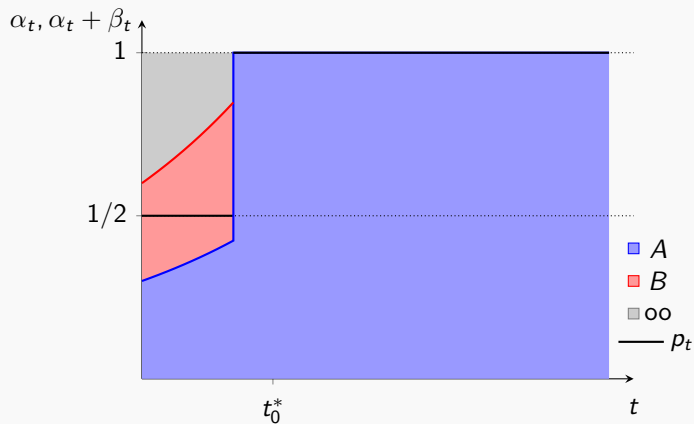


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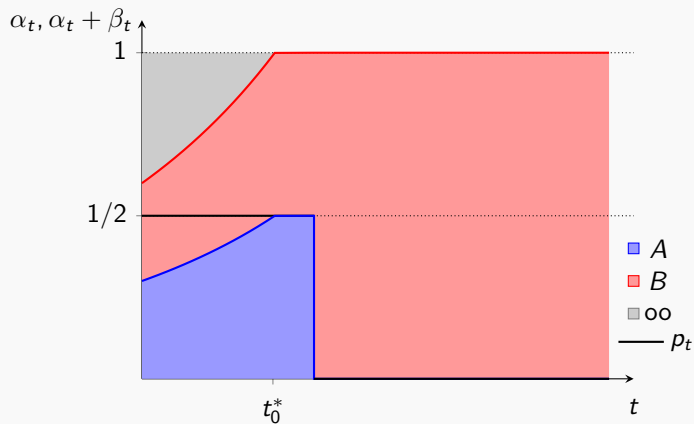


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The Optimal Policy in a Special Case

Assume $p^0 = 1/2$ and $\omega_A = \omega_B = 0$.

- ▶ If $\omega_A = 0$, then $\alpha_t \equiv \beta_t$ for all $t \geq 0$.
- ▶ The exploration pattern is symmetric.
- ▶ Conditional on either product being superior, the expected time to discovery is the same.

The Optimal Policy in a Special Case

Assume $p^0 = 1/2$ and $c > \omega_A > \omega_B = 0$.

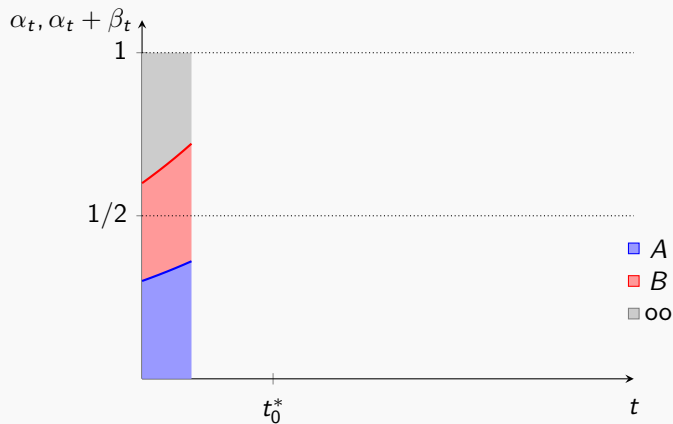


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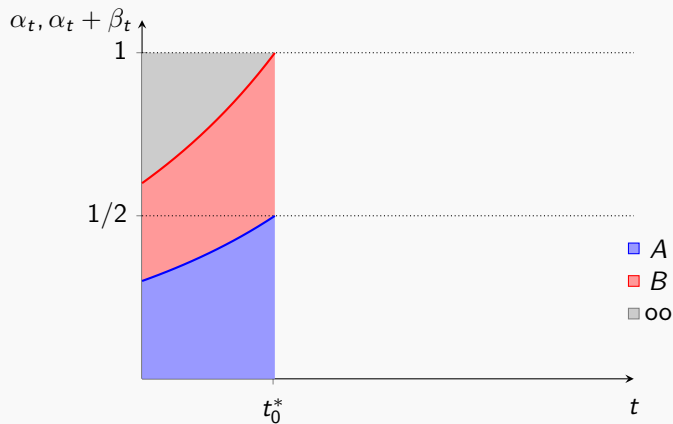


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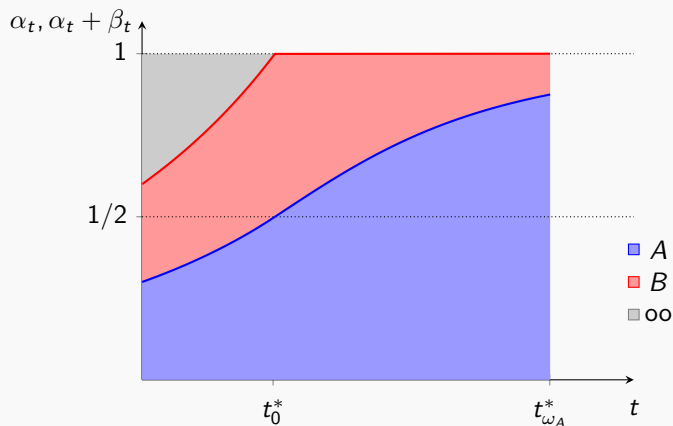


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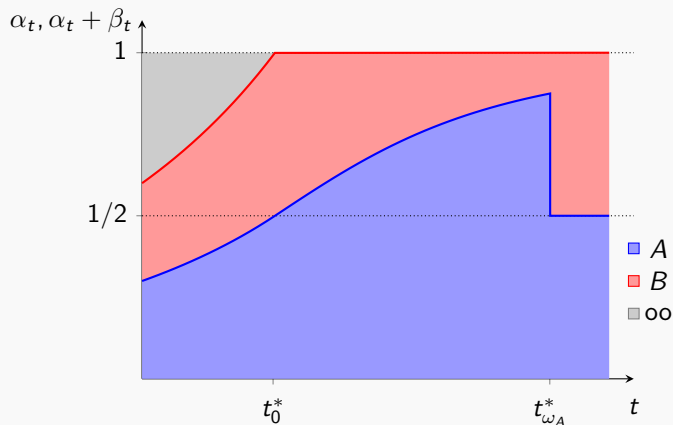


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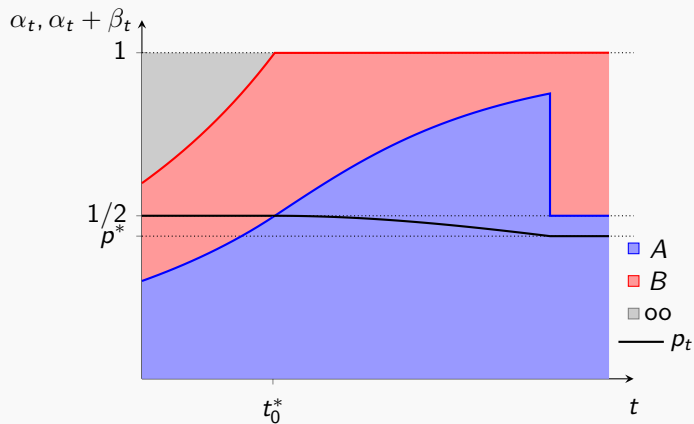


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The Optimal Policy in a Special Case

- ▶ If $\omega_A > 0$, then $\alpha_t > \beta_t$ for all $t_0^* < t < t_{\omega_A}^*$.
- ▶ The designer does not recommend product A to more consumers because she is more optimistic about its quality.
- ▶ Rather, allocating more consumers to product A today increases the likelihood of receiving conclusive feedback about it and strengthens the designer's perceived expertise, i.e., relaxes future incentive constraints when recommending product A .

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Optimal Recommendation Policy

Let

$$\underline{\ell} := \frac{c - \omega_A}{1 + \omega_A - \omega_B}$$

$$\ell^* := \frac{p^*}{1 - p^*}$$

$$\bar{\ell} := \frac{1 + \omega_B - \omega_A}{c - \omega_B}$$

Optimal Recommendation Policy

Assume $\bar{\beta}(\ell_0, B_0) < \bar{\alpha}(\ell_0, B_0) < 1$, $\bar{\alpha}(\ell_0, B_0) + \bar{\beta}(\ell_0, B_0) < 1$, and $\ell_0 \in [1, \bar{\ell}]$.

Theorem

The optimal recommendation policy has the following configurations:

$$\alpha_t = \begin{cases} \bar{\alpha}(\ell_t, B_t) & 0 \leq t \leq t^{(2)} \\ \end{cases} \quad \beta_t = \begin{cases} \bar{\beta}(\ell_t, B_t) & 0 \leq t \leq t^{(1)} \\ \end{cases}$$

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Theorem

The optimal recommendation policy has the following configurations:

$$\alpha_t = \begin{cases} \bar{\alpha}(\ell_t, B_t) & 0 \leq t \leq t^{(2)} \\ 1 - \bar{\beta}(\ell_t, B_t) & t^{(2)} < t \leq t^{(3)} \\ < 1 - \bar{\beta}(\ell_t, B_t) & t^{(3)} < t \leq t^{(4)} \\ 1 - \bar{\beta}(\ell_t, B_t) & t^{(4)} < t \leq t^* \\ \frac{1}{2} & t > t^* \end{cases} \quad \beta_t = \begin{cases} \bar{\beta}(\ell_t, B_t) & 0 \leq t \leq t^{(1)} \\ 1 - \bar{\alpha}(\ell_t, B_t) & t^{(1)} < t \leq t^{(2)} \\ \bar{\beta}(\ell_t, B_t) & t^{(2)} < t \leq t^* \\ \frac{1}{2} & t > t^* \end{cases}$$

One or more of the inequalities $t^{(1)} \leq t^{(2)} \leq t^{(3)} \leq t^{(4)} \leq t^*$ may hold as equalities. Further, $\ell_t \equiv \ell^*$ for $t \geq t^*$. If $t^{(1)} < t^{(2)}$, then $\ell_{t^{(1)}} > \ell^*$.

Optimal Recommendation Policy

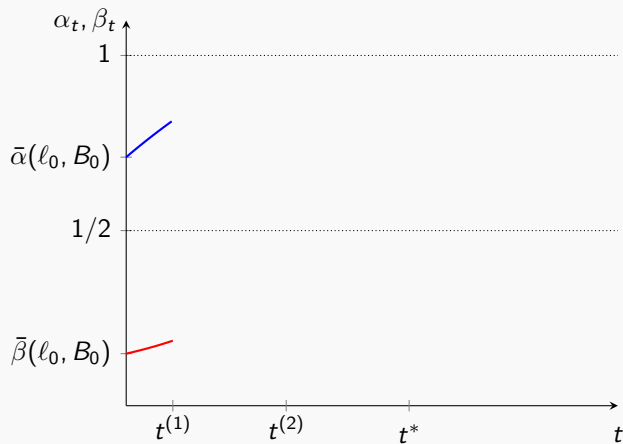


Figure: $c = 17/25$, $\ell_0 = 25/17$, $B_0 = 19/25$, $\omega_A = \omega_B = 0$.

Optimal Recommendation Policy

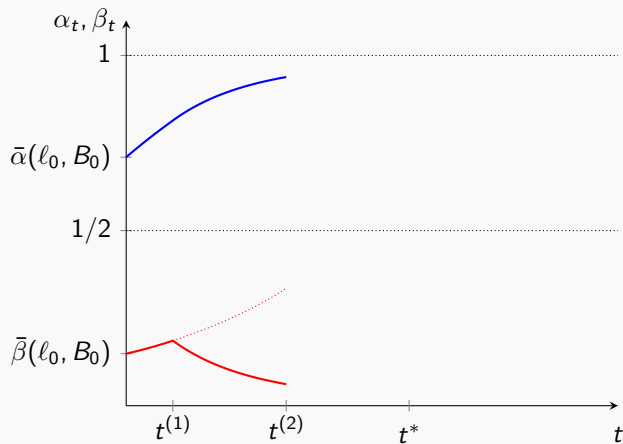


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Optimal Recommendation Policy

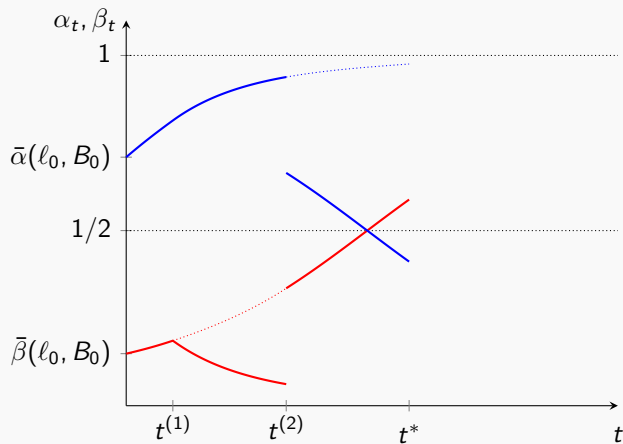


Figure: $c = 17/25$, $\ell_0 = 25/17$, $B_0 = 19/25$, $\omega_A = \omega_B = 0$.

Optimal Recommendation Policy

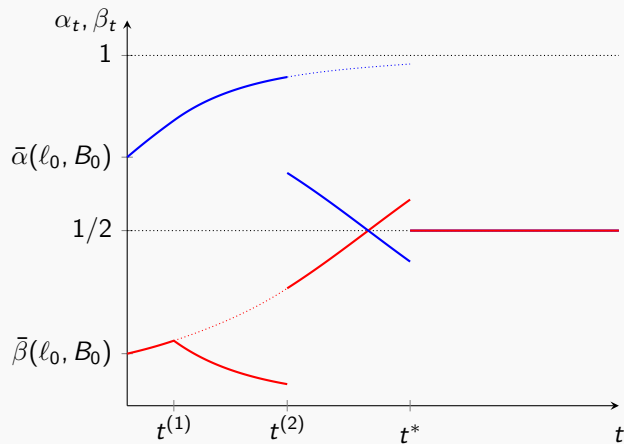


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Optimal Recommendation Policy

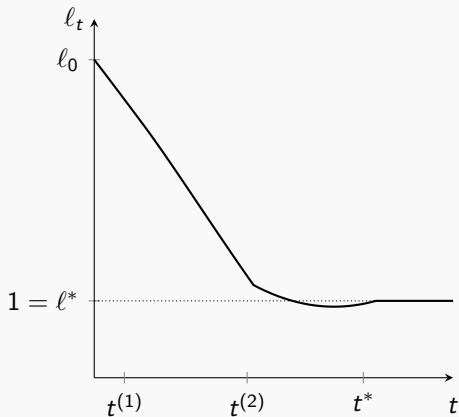


Figure: $c = 17/25$, $\ell_0 = 25/17$, $B_0 = 19/25$, $\omega_A = \omega_B = 0$.

Exploration Builds Credibility

- ▶ When deciding which product to recommend, the designer trades off current exploitation against future ability to experiment.
- ▶ Recommending a product today increases future credibility when recommending that same product.
- ▶ At $t^{(2)}$, the designer starts recommending product B to as many agents as possible, even though $\ell_{t^{(2)}} > \ell^*$.

Exploration Builds Credibility

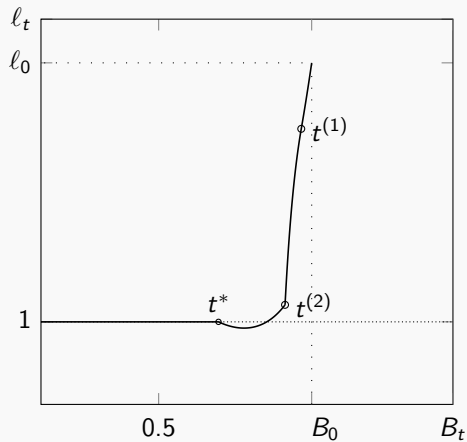


Figure: $c = 17/25$, $\ell_0 = 25/17$, $B_0 = 19/25$, $\omega_A = \omega_B = 0$.

The Role of ω_A and ω_B

- ▶ The structure of the optimal recommendation policy does not depend on the magnitude of ω_A and ω_B .
- ▶ However, the duration of the different phases does depend on ω_A and ω_B .
- ▶ Intuitively, ω_A and ω_B shift the designer's incentives to experiment with each product.

The Role of ω_A and ω_B

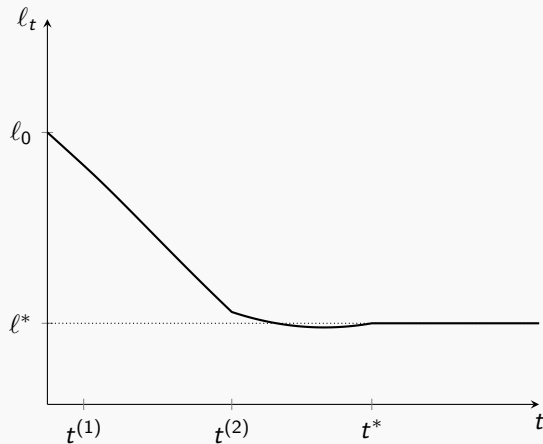


Figure: $c = 17/25$, $l_0 = 25/17$, $B_0 = 3/4$, $\omega_A = 0$, $\omega_B = 0$.

- Higher ω_B , lower $t^{(2)}$ and higher t^* .

The Role of ω_A and ω_B

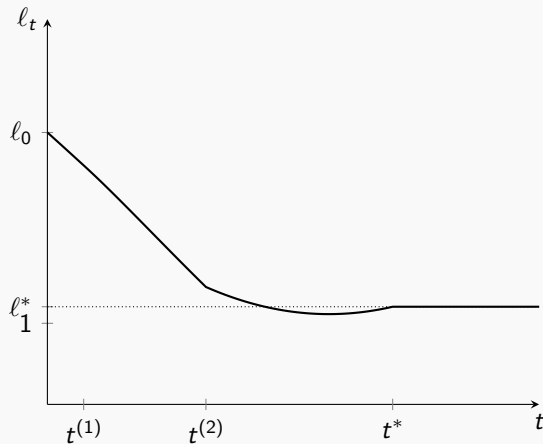


Figure: $c = 17/25$, $l_0 = 25/17$, $B_0 = 3/4$, $\omega_A = 0$, $\omega_B = 1/50$.

- Higher ω_B , lower $t^{(2)}$ and higher t^* .

The Role of ω_A and ω_B

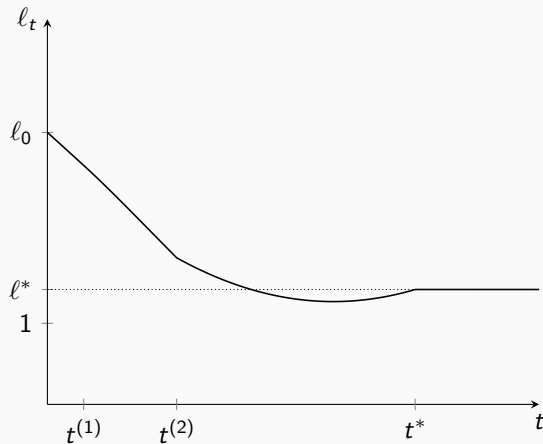


Figure: $c = 17/25$, $l_0 = 25/17$, $B_0 = 3/4$, $\omega_A = 0$, $\omega_B = 1/25$.

- Higher ω_B , lower $t^{(2)}$ and higher t^* .

The Role of ω_A and ω_B

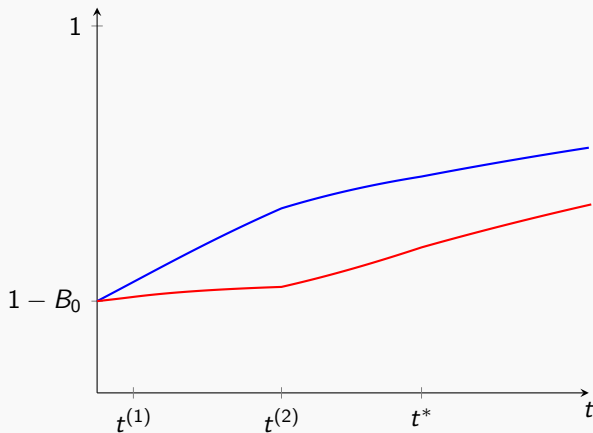


Figure: Probability that the designer has discovered $\theta = A$ (blue) or $\theta = B$ (red) by time t .
 $c = 17/25$, $\ell_0 = 25/17$, $B_0 = 3/4$, $\omega_A = 0$, $\omega_B = 0$.

- ω_A and ω_B do not change the structure of the policy, but they change the speed of discovery.

The Role of ω_A and ω_B

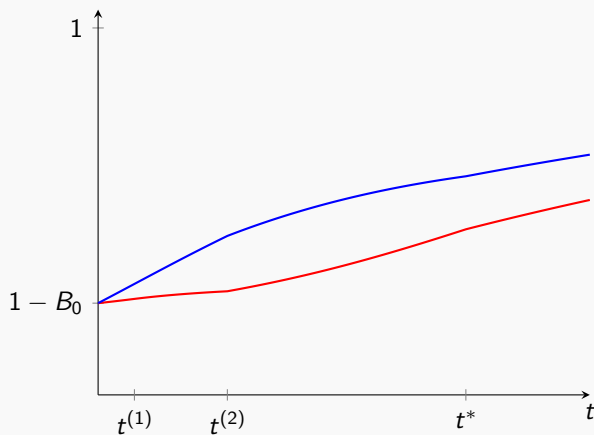


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Outline

1. Model
2. Dynamic Incentive Problem
3. The Optimal Policy in a Special Case
4. Main Result
5. Implications for Self-Preferencing
6. Conclusion

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1. Model
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Self-Preferencing

- ▶ Interpreting product A as the platform's own product, I investigate self-preferencing by assuming $\omega_B = 0$ and $0 \leq \omega_A < 1 - c$.
- ▶ I focus on the special case of symmetric products, $\ell_0 = 1$.
- ▶ How does self-preferencing qualitatively change the optimal policy?

Self-Preferencing

Proposition

Assume $\ell_0 = 1$ and $\omega_B = 0$.

1. If $\omega_A = 0$, the optimal policy involves $\alpha_t \equiv \beta_t < 1/2$ for all $t < t_0^*$, and $\ell_t \equiv 1$ for all $t \geq 0$.
2. If $\omega_A \in (0, 1 - c)$, the optimal policy has the following configuration:

$$\alpha_t = \begin{cases} \bar{\alpha}(\ell_t, B_t) & 0 \leq t \leq t_{\omega_A}^* \\ 1/2 & t > t_{\omega_A}^* \end{cases} \quad \beta_t = \begin{cases} \bar{\beta}(\ell_t, B_t) & 0 \leq t \leq t_{\omega_A}^{(1)} = t_0^* \\ 1 - \bar{\alpha}(\ell_t, B_t) & t_{\omega_A}^{(1)} < t \leq t_{\omega_A}^* \\ 1/2 & t > t_{\omega_A}^*. \end{cases}$$

Further, for any $\omega_A > 0$, $\ell_t \geq \ell_{\omega_A}^*$ for all $t \geq 0$.

Self-Preferencing

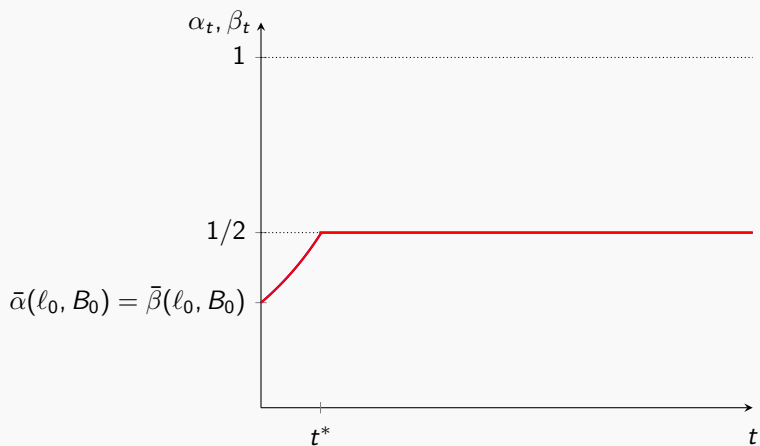


Figure: Optimal Policy. $c = 5/8$, $\ell_0 = 1$, $B_0 = 5/6$, $\omega_A = 0$, $\omega_B = 0$.

Self-Preferencing

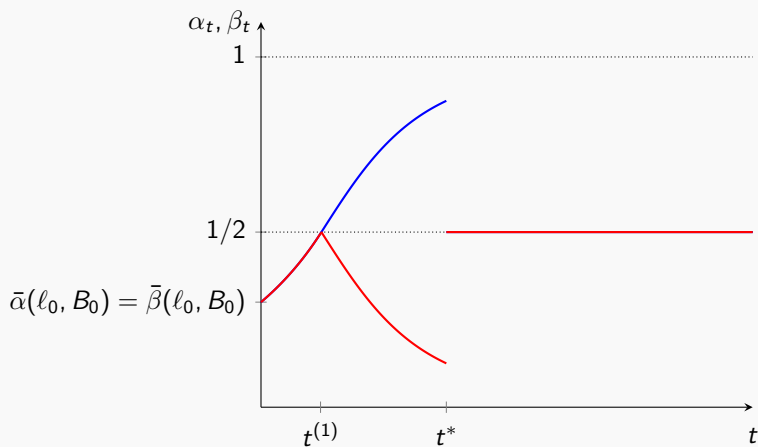


Figure: Optimal Policy. $c = 5/8$, $\ell_0 = 1$, $B_0 = 5/6$, $\omega_A = 1/8$, $\omega_B = 0$.

Self-Preferencing

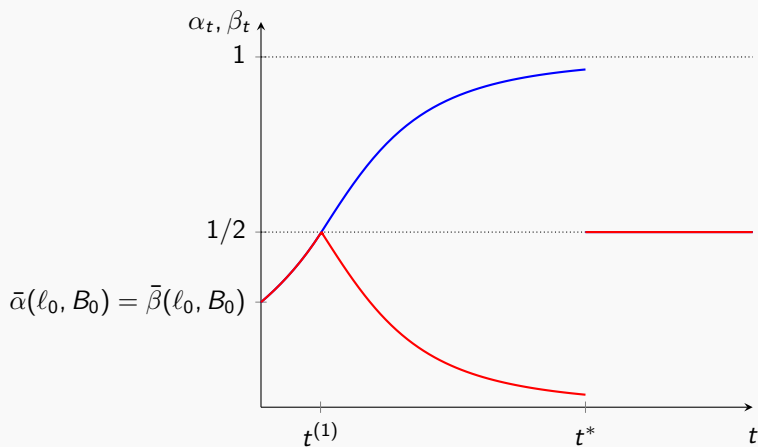


Figure: Optimal Policy. $c = 5/8$, $\ell_0 = 1$, $B_0 = 5/6$, $\omega_A = 3/8$, $\omega_B = 0$.

Self-Preferencing

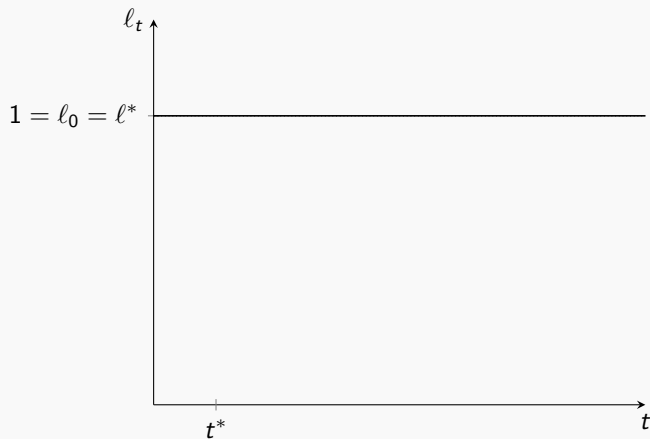


Figure: Belief Trajectory. $c = 5/8$, $\ell_0 = 1$, $B_0 = 5/6$, $\omega_A = 0$, $\omega_B = 0$.

Self-Preferencing

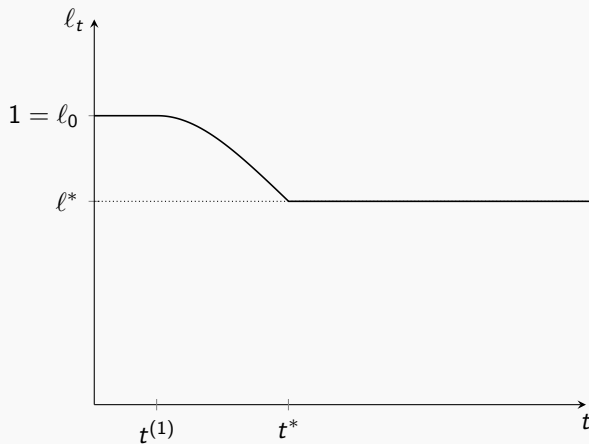


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Self-Preferencing

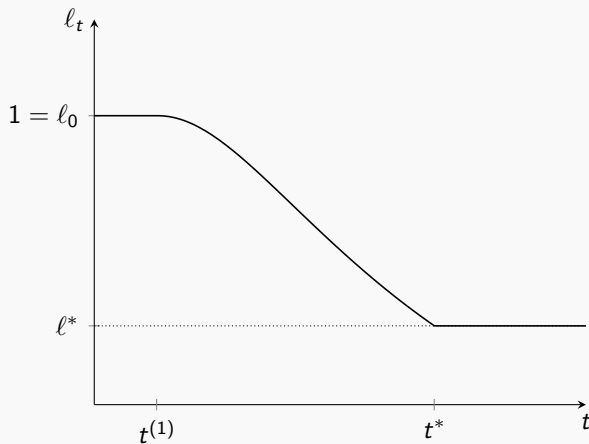


Figure: Belief Trajectory. $c = 5/8$, $\ell_0 = 1$, $B_0 = 5/6$, $\omega_A = 3/8$, $\omega_B = 0$.

Self-Preferencing

- ▶ Self-preferencing leads the platform to recommend its own product more often (Farronato, Fradkin, and MacKay, 2023).
- ▶ This slows down learning and delays the discovery of superior third-party products.
- ▶ Without access to the platform's private information, this distortion is hard to detect.

Detecting Self-Preferencing: Delay

- ▶ When the platform engages in self-preferencing, it explores the third-party product less aggressively: even when the third-party product is superior, the platform learns this later.
- ▶ In the meantime, the platform continues to recommend its own product more often, so its expected sales remain higher throughout.
- ▶ More broadly, the model shows that self-preferencing affects not only rankings, but also learning dynamics.

▶ conclusions

Detecting Self-Preferencing: Delay

Corollary

Assume $\ell_0 = 1$ and $\omega_B = 0$. Discovery times are ordered by first-order stochastic dominance: as ω_A increases, the platform discovers that B is superior stochastically later and discovers that A is superior stochastically earlier.

► conclusions

Detecting Self-Preferencing: Delay

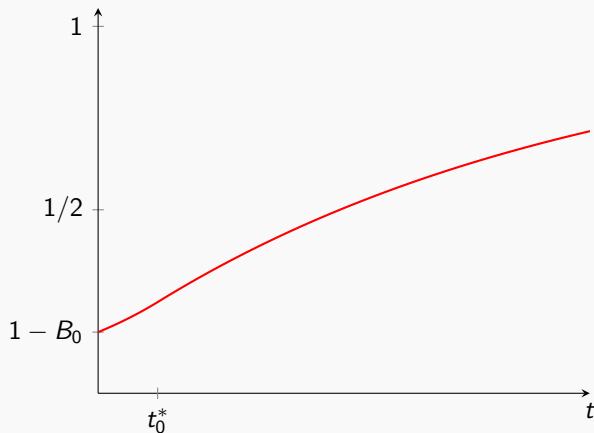


Figure: Probability that the designer has discovered $\theta = A$ (blue) or $\theta = B$ (red) by time t .
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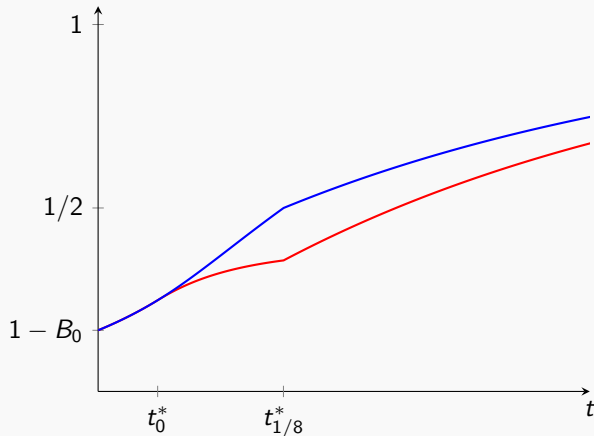


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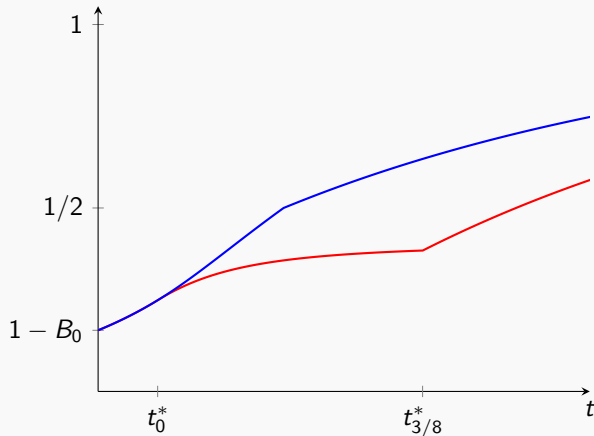


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Detecting Self-Preferencing: Market Shares

Corollary

Assume $\ell_0 = 1$ and $\omega_B = 0$. For any $t \geq 0$, the ex-ante expected sales of the platform at any time t are increasing in ω_A .

The platform's average market share across markets in which it acts as both seller and recommender may serve as a simple rule-of-thumb indicator for detecting self-preferencing.

Detecting Self-Preferencing: Market Shares

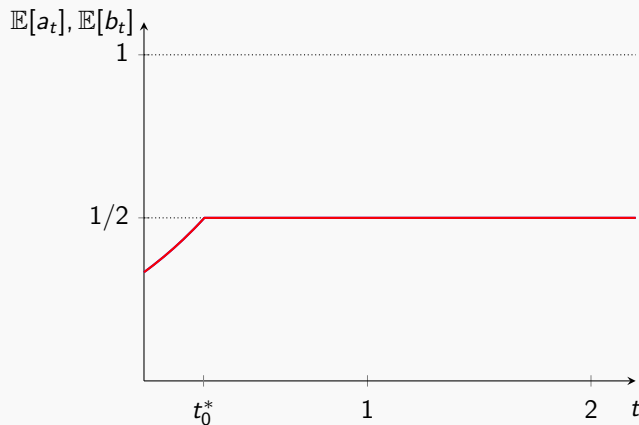


Figure: Expected market share of product A (blue) and product B (red). $c = 5/8$, $\ell_0 = 1$, $B_0 = 5/6$, $\omega_A = 0$, $\omega_B = 0$.

Detecting Self-Preferencing: Market Shares

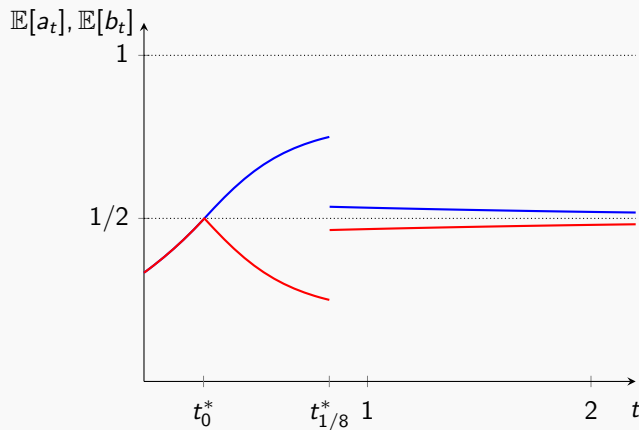


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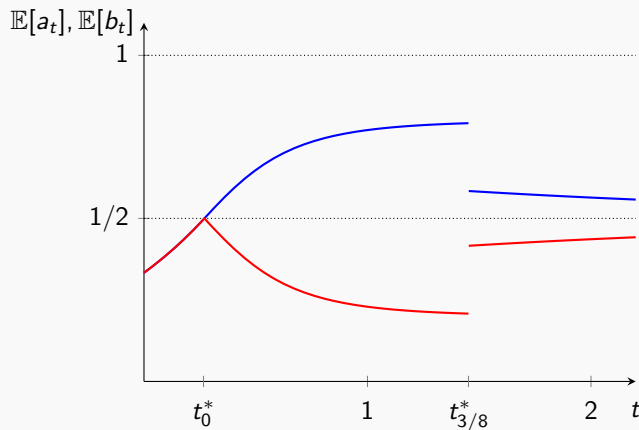


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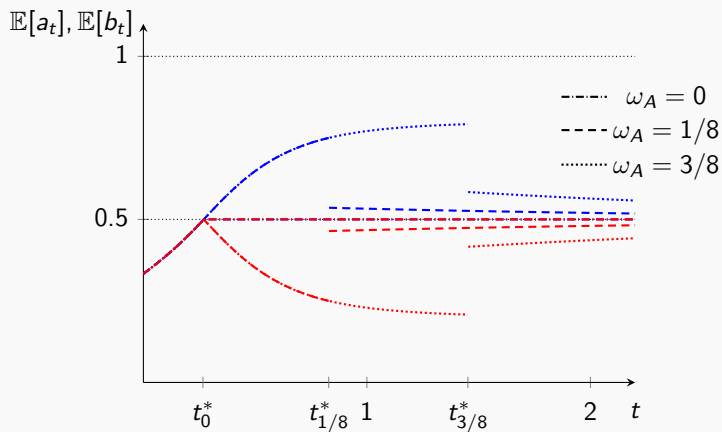


Figure: Expected market share of product A (blue) and product B (red). $c = 5/8$, $\ell_0 = 1$, $B_0 = 5/6$, $\omega_B = 0$.

Detecting Self-Preferencing: Market Shares

- ▶ In the Corollary, the assumption $\ell_0 = 1$ guarantees that the structure of the optimal policy is unchanged across different values of ω_A .
- ▶ However, the comparative statics result holds for any prior, provided that $t^{(2)} = t^*$ for $\omega_A = 0$.

Detecting Self-Preferencing: Market Shares

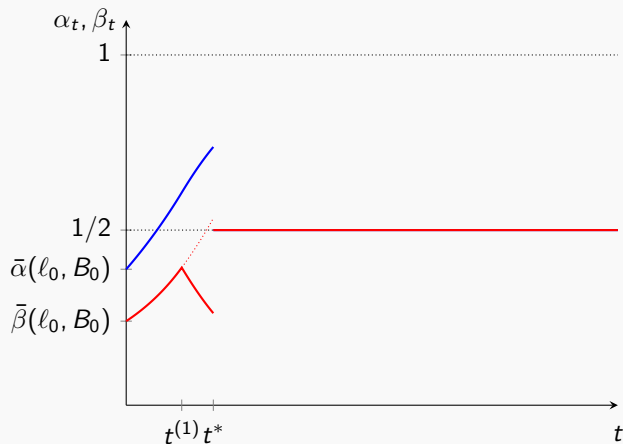


Figure: Optimal Policy. $c = 5/8$, $\ell_0 = 11/10$, $B_0 = 5/6$, $\omega_A = 0$, $\omega_B = 0$.

Detecting Self-Preferencing: Market Shares

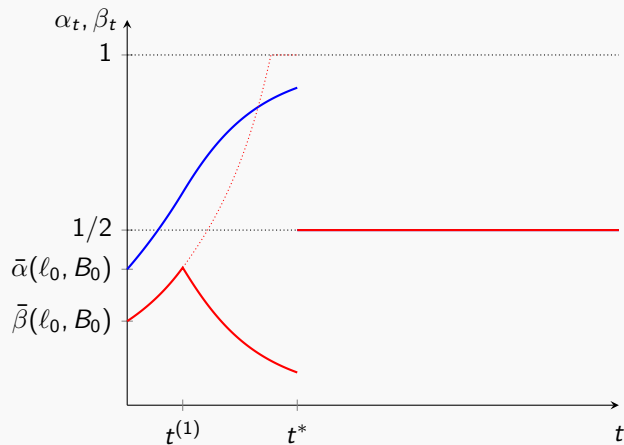


Figure: Optimal Policy. $c = 5/8$, $\ell_0 = 11/10$, $B_0 = 5/6$, $\omega_A = 1/8$, $\omega_B = 0$.

Detecting Self-Preferencing: Market Shares

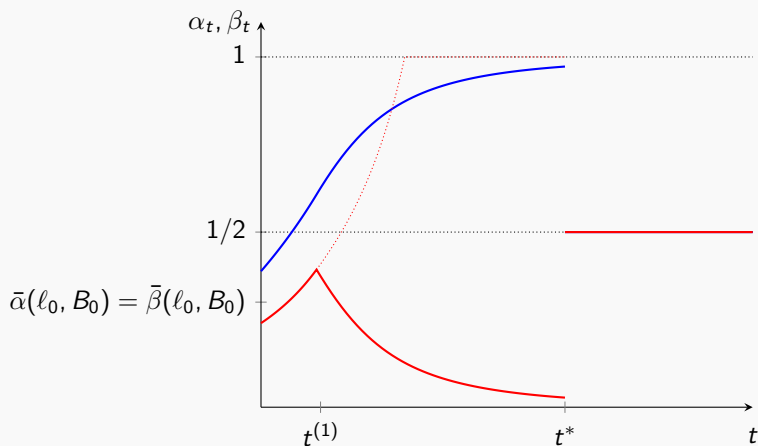


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Detecting Self-Preferencing: Market Shares

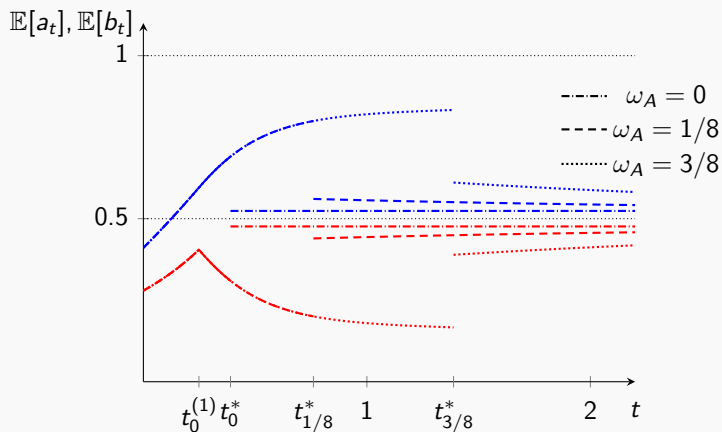


Figure: Expected market share of product A (blue) and product B (red). $c = 5/8$, $\ell_0 = 11/10$, $B_0 = 5/6$, $\omega_B = 0$.

Conclusions

- ▶ I study a recommender system that mediates social learning and builds credibility over time through its recommendation.
- ▶ The optimal policy is characterized by distinct phases reflecting a trade-off between current exploitation and future experimentation.
- ▶ Self-preferencing affects not only rankings and market shares, but the speed of product discovery: platforms over-explore their own products and delay the discovery of superior third-party options.
- ▶ Persistent asymmetries in market shares or exploration paths can signal self-preferencing.

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- ▶ Persistent asymmetries in market shares or exploration paths can signal self-preferencing.
- ▶ If a platform could set prices and recommendations, how would it trade off these instruments to maximize its profits?
- ▶ How does the optimal recommendation policy change if agents are heterogeneous?

▸ Literature

▸ Public Persuasion

Literature

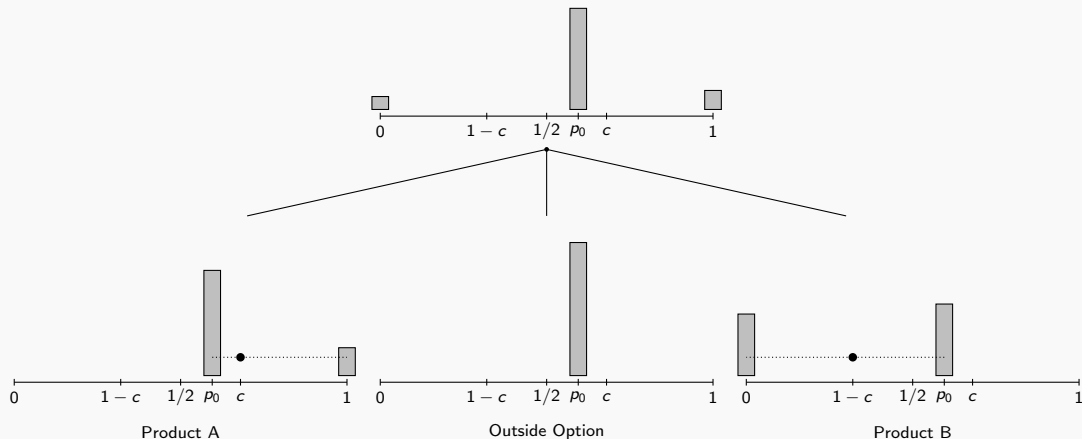
- ▶ **Design for Social Learning.** Kremer, Mansour, and Perry (2014) and Che and Hörner (2018). Only one uncertain product and benevolent designer.
- ▶ **Theoretical IO literature on steering.** Hagiu and Jullien (2011) and following. The platform has superior information at the outset.
- ▶ **Persuasion Literature.** Ely (2017) and Renault, Solan, and Vieille (2017). The optimal policy is greedy because the agents only have two options.

Public Persuasion

- ▶ Under public recommendations, the designer solves a dynamic programming problem with the agent's second-order belief μ as state variable.

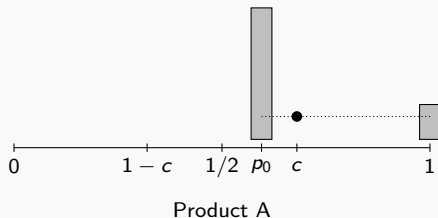
Public Persuasion

- ▶ Under public recommendations, the designer solves a dynamic programming problem with the agent's second-order belief μ as state variable.
- ▶ Assume $\omega_A = \omega_B = 0$.
- ▶ The optimal policy involves a three-way split of the initial belief.



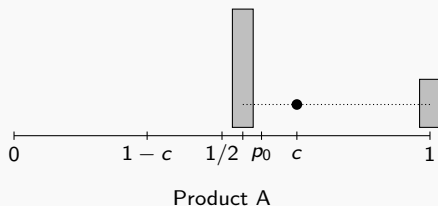
Public Persuasion

- ▶ Following a recommendation of product A, the designer continues recommending A as long as



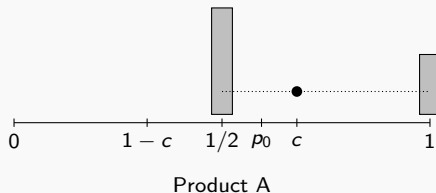
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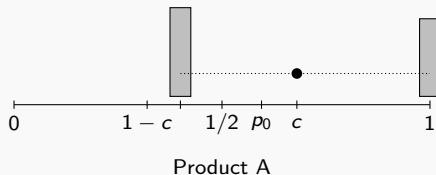
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- ▶ Following a recommendation of product A, the designer continues recommending A as long as



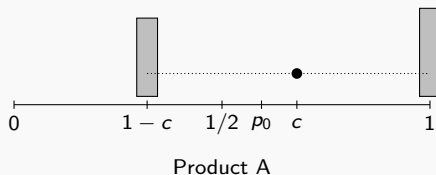
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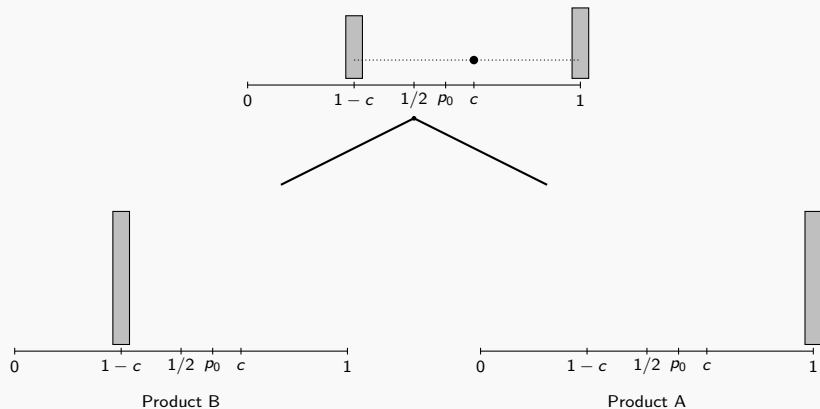
Public Persuasion

- ▶ Following a recommendation of product A , the designer continues recommending A as long as $\min \text{supp } \mu > 1 - c$.



Public Persuasion

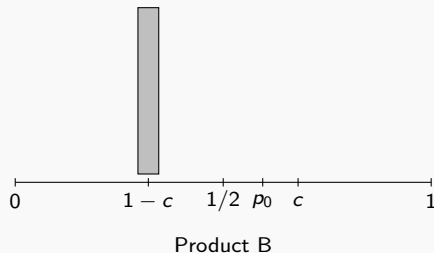
- ▶ Following a recommendation of product A , the designer continues recommending A as long as $\min \text{supp } \mu > 1 - c$.
- ▶ If $\min \text{supp } \mu > 1 - c$, the designer “splits” the second-order belief:



- ▶ The product A recommendation, i.e., the posterior δ_1 , is absorbing.
- ▶ Following a product B recommendation, the designer continues recommending B as long as $\max \text{supp } \mu < c$.

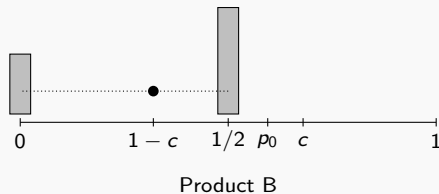
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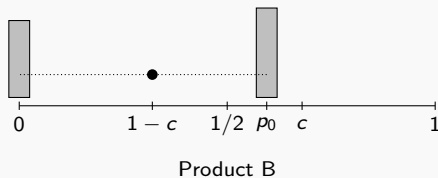
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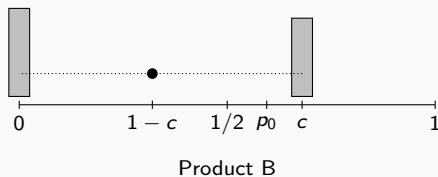
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Public Persuasion

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- ▶ Following a product B recommendation, the designer continues recommending B as long as $\max \text{supp } \mu < c$.



Public Persuasion

- ▶ The designer alternates between recommending A and B before eventually settling on a single recommendation.
- ▶ If $\omega_A \neq 0$ and/or $\omega_B \neq 0$, the designer continues recommending A (B) as long as $\min \text{supp } \mu > \underline{p}$ ($\max \text{supp } \mu < \bar{p}$). If $\min \text{supp } \mu = \underline{p}$ ($\max \text{supp } \mu = \bar{p}$), she “splits” the second-order belief between δ_1 and $\delta_{\underline{p}}$, ($\delta_{\bar{p}}$ and δ_0) recommending product A and product B, respectively.
- ▶ The cutoffs $\underline{p}$ and $\bar{p}$ satisfy

$$\underline{p} = \begin{cases} 1 - c, & \text{if } \omega_A \leq \omega_B \\ \in [0, 1 - c), & \text{if } \omega_A > \omega_B, \end{cases} \quad \bar{p} = \begin{cases} c, & \text{if } \omega_A \geq \omega_B \\ \in (c, 1], & \text{if } \omega_A < \omega_B, \end{cases}$$

and

$$(\bar{p}(2\underline{p} - 1) - \underline{p} + 1)(\omega_A - \omega_B) = 1 - \bar{p} - \underline{p}.$$

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